review of the terminology concerning subatomic particles.

all particle are either..

	Bosons (integer spin)	Fermions (half-integer spin)	cf. spin-statistic theorem.
$m=0$	photons. gluons $(s=1)$	...	
	gravitons $(s=2)$		
$m \neq 0$	...	electron $(s=\frac{1}{2})$	

[1] Folland, G.B. Quantum field theory: a tourist guide for mathematicians.

[2] Lee, John M. Introduction to Smooth Manifolds.

Single-particle space.

[1] relativistic invariance implies that the state space $\mathcal{H}$ must come equipped with a projective unitary representation of the group $\mathbb{R}^{1,3} \rtimes SO^{\uparrow}(1,3)$.

Moreover, every projective unitary representation of $\mathbb{R}^{1,3} \rtimes SO^{\uparrow}(1,3)$ comes from genuine unitary representation of its universal cover $\mathbb{R}^{1,3} \rtimes SL(2,\mathbb{C})$

(We shall assume that this representation is irreducible, which means that the particle is in some sense "elementary".)

In [1], we can study the irreducible unitary representation of $\mathbb{R}^{1,3} \rtimes SL(2,\mathbb{C})$.

And that covers general cases $(s \geq 0, m \geq 0)$. But I will follow our textbook.

For a model for electrons, we set

m = (the rest mass of the electron), $\mathcal{H} = L^2(X_m, d\lambda_m, \mathbb{C}^2)$

$\mathbb{R}^{1,3} \rtimes SO^{\uparrow}(1,3)$
$\parallel$

$U : \mathcal{P} \to \{U(a,A) : \mathcal{H} \circlearrowleft ,\ (a,A) \in \mathcal{P}\}$

Q. inner product ?, the repn.^{U} of the Poincaré group, $\mathcal{P}$?

To answer, we modify the (5) postulate in Lec. 11.

"If an observer in a given coordinate system observes the system to be in state ψ,

then an observer in a new coordinate system obtained by the action of (a,A) on $\mathbb{R}^{1,3}$

observes the state of the same system as $U(a,A)\psi$."

Here, U is a representation $\rightarrow$ $U((a,A)(b,B)) = U(a,A)\,U(b,B)$

Note expectations of observables under a state ψ are invariant under multiplication

of the state by a phase $\alpha \in \mathbb{C}$, $|\alpha| = 1$, (See lec 3)

it doesn't violate the postulate (4).

2.

Thus it suffices that U is unitary and a projective representation,

meaning that

$$U((a,A)(b,B)) = r(a,A,b,B)\,U(a,A)\,U(b,B), \quad r(a,A,b,B) \in \{z \in \mathbb{C} : |z| = 1\}.$$

<u>Recall</u> p.38 $\mathcal{P} = \mathbb{R}^{1,3} \rtimes SO^{\uparrow}(1,3)$.

The Lorentz group $O(1,3)$: the group of all linear transformation of $\mathbb{R}^{1,3}$ that preserve the Lorentz Inner product :

Conjugate transpose.

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$$A \in O(1,3) \Leftrightarrow (Ax)_\mu (Ay)^\mu = x_\mu y^\mu \text{ for all } x, y \Leftrightarrow A^\dagger g A = A$$

- $O(1,3)$ has 4 connected components, which are determined by values of two homomorphisms $O(1,3) \to \{\pm 1\}$ $\begin{cases} A \mapsto \det A \\ A \mapsto \mathrm{sgn}(A^0_0) \end{cases}$

$\ker[A \mapsto \det(A)] = SO(1,3)$: the special Lorentz group

$\ker[A \mapsto \mathrm{sgn}(A^0_0)] = O^{\uparrow}(1,3)$: the orthochoronous Lorentz group.

$SO^{\uparrow}(1,3) = SO(1,3) \cap O^{\uparrow}(1,3)$: the restricted Lorentz group

; the connected component of the identity of $O(1,3)$.

25.2 $SL(2,\mathbb{C})$: 2x2 complex matrices of determinant 1.

$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$. $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$: Pauli matrices.

form a basis for the space H of 2x2 Hermitian matrices.

Consider a bijection $M : \mathbb{R}^{1,3} \to H$

$$x \in \mathbb{R}^{1,3} \longleftrightarrow M(x) = \begin{pmatrix} x^0 + x^3 & x^1 - ix^2 \\ x^1 + ix^2 & x^0 - x^3 \end{pmatrix}$$

Note. $\det(M(x)) = x^2 \leftarrow$ Minkowski

Every $A \in SL(2,\mathbb{C})$ acts on H by the map $X \mapsto AXA^\dagger$, we denote the corresponding action on $\mathbb{R}^{1,3}$ by $k(A)$:

$$M(k(A)x) = AM(x)A^\dagger$$

$|A|=1$

Note $\det(AMS) = \det(M(x)) = x^2$ & $k(A)$: linear for all $A \in SL(2,\mathbb{C})$.

Thus $k: SL(2,\mathbb{C}) \longrightarrow O(1,3)$. (norm preserving.)

$$\det(k(A)x) = \det(M(k(A)x)) = \det(AM(x)A^{+}) = x^2.$$

facts.

✱ k is a 2-to-1 surjective homomorphism from $SL(2,\mathbb{C})$ onto $SO^{\uparrow}(1,3)$.

(Sketch) the differential of k at the identity, which takes $S \in sl(2,\mathbb{C})$ to

the map $M(x) \mapsto SM(x) + M(x)S^{+}$, is an isomorphism of Lie algebras ([1] p.11)

Since $SL(2,\mathbb{C})$ is connected, k factor through the connected component

$SO^{\uparrow}(1,3)$ of $O(1,3)$. Note that its kernel is $\{I,-I\}$. Thus k is

a Lie group homomorphism betw two connected Lie groups. Thus by

theorem ([2] theorem 2.31), k is a double covering of $SO^{\uparrow}(1,3)$ by $SL(2,\mathbb{C})$.

(This proof is based on [1].)

· "$k(A) = k(B)$" iff "$A=B$ or $A=-B$". (We will write $A = \pm B$)

✱✱ $k(A^{+}) = k(A)^{t}$ (omit the proof)

By ✱, $\exists \rho$ such that $k \circ \rho$ (as below) is the identity on $SO^{\uparrow}(1,3)$.

$$SO^{\uparrow}(1,3) \xrightarrow{\exists \rho} SL(2,\mathbb{C}) \xrightarrow{k} SO^{\uparrow}(1,3)$$

$k(A)$ A $-A$ $k(A)$

Note $\begin{cases} \rho(AB) = \pm \rho(A)\rho(B) \\ \rho(A^{+}) = \rho(A^{+}) \quad (✱✱) \end{cases}$

25.3

Assume that there is a such ρ so that the projective repn. of $SO^{\uparrow}(1,3)$ in $\mathcal{H} = L^2(X_m, d\lambda_m, \mathbb{C}^2)$ is given by

$$(U(a,A)\psi)(p) = e^{i(a,p)}\rho(A)\psi(A^{-1}p). \qquad (\text{fixed } (a,A) \in \mathcal{P})$$

- It is a projective repn.

$$U(a,A)\underbrace{(U(b,B)\psi)}_{\in\mathcal{H}}(p) = e^{i(a,p)}\rho(A)\,(U(b,B)\psi)(A^{-1}p) \qquad \tilde{\psi}(\cdot) := \psi(A^{-1}\cdot)$$

$$= e^{i(a,p)}e^{i(b,p)}\rho(A)\rho(B)\psi(B^{-1}A^{-1}p)$$

$(a,A)(b,B) = (a+Ab, AB)$ (p.43)

$$= \pm e^{i(b-Ab,p)}\,(U(a,A)U(b,B)\psi)(p)$$

Minkowski.

//

- To prove that $U(a,A)$ is unitary, we will define the inner product on $\mathcal{H}$.

Denote $p^* = (m,0,0,0) \in X_m$. For any $p \in X_m$, $\exists$ maps taking p^* to p.

($\because$ The action of $SO^{\uparrow}(1,3)$ on X_m is transitive.

$\uparrow$ In $X_m = \{ p \in \mathbb{R}^{1,3} : p^2 = m^2, p^0 > 0 \}$, every element has the same mass, so that one can get a hyperbola btw any two points in X_m.)

If A, B are any such maps, then

$$A = BR, \quad R = \begin{pmatrix} 1 & 0 \\ 0 & Q \end{pmatrix}, \quad \exists Q \in SO(3)$$

Spatial rotation.

($\because B \circ A^{-1}(p^*) = p^*$; fix time. $\left[\begin{array}{c|c} 1 & \\ \hline & ////\end{array} \right]\begin{bmatrix} m \\ \vdots \end{bmatrix} = \begin{pmatrix} m \\ \vdots \end{pmatrix}$)

$\therefore \forall p \in X_m$, $\exists$ a unique positive-definite $V_p \in SL(2,\mathbb{C})$ such that $\kappa(V_p)(p^*) = p$.

(uniqueness is followed from the positive-definiteness)

$A \in SL(2,\mathbb{C})$
$\rightarrow A = PU \rightarrow SU(2)$
$\hookrightarrow$ positive defin. Hermitian $|P|=1 \rightarrow SL(2,\mathbb{C})$

If $AA^{\dagger} = BB^{\dagger} \Rightarrow I = A^{-1}B(A^{-1}B)^{\dagger}$

p.113.

$$\begin{array}{ccc} SU(2) & \hookrightarrow & SL_2(\mathbb{C}) \\ \kappa|_{SU(2)} \downarrow & & \downarrow \kappa \\ SO(3) & \hookrightarrow & SO^{\uparrow}(1,3) \end{array}$$

<u>uniqueness</u>

$\kappa(A')(p^*) = p$ 이라면

$\rightarrow A' = AU'$, $U' \in SU(2)$

이므로 같은 B를 얻는다.

take $A \in SL_2(\mathbb{C})$ s.t. $\kappa(A)(p^*) = p$.

$A = BU$: polar decomposition.

$SU(2)$ fixes p^*, we take $B = V_p$

$\kappa(BU)(p^*) = \kappa(B)(p^*) = p$.

Now we define the inner product:

$$(\psi, \varphi) = \int_{X_m} \psi(p)^\dagger V_p^{-2} \varphi(p)\, d\lambda_m(p), \qquad \psi, \varphi \in \underbrace{L^2(X_m, d\lambda_m, \mathbb{C}^2)}$$

↳ Hilbert space under this $(\cdot,\cdot)$

Claim: $U(a,A)$ is unitary ; $(U(a,A)\psi, U(a,A)\varphi) = (\psi, \varphi)$

To this end, Note

$$(U(a,A)\psi)(p)^\dagger V_p^{-2} (U(a,A)\varphi)(p) = \psi(A^{-1}p)^\dagger \rho(A)^\dagger V_p^{-2} \rho(A) \varphi(A^{-1}p). \qquad \text{by Construction.}$$

$$(U(a,A)\psi, U(a,A)\varphi) = \int_{X_m} \psi(p)^\dagger \underbrace{\rho(A)^\dagger V_{Ap}^{-2} \rho(A)}_{\overset{?}{=} V_p^{-2}} \varphi(p)\, d\lambda_m(p).$$

$\int_{X_m = \{p^2 = m^2\}} d\lambda_m(p) f(p) = \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} f(\omega_p, p)$

$A \in SO^\uparrow(1,3)$

Lemma 25.1.

$$\rho(A)^\dagger V_{Ap}^{-2} \rho(A) = V_p^{-2}$$

proof.

$$k(V_{Ap}^{-1} \rho(A)) = k(V_{Ap}^{-1}) \underbrace{k(\rho(A))}_{=1} = k(V_{Ap})^{-1} A. \qquad (\because k: \text{homomorphism})$$

$$A^{-1} k(V_{Ap})(p^*) = A^{-1}Ap = p \qquad (k(V_{Ap})(p^*) = p)$$

$\rightarrow$ $A^{-1} k(V_{Ap}) = k(V_p) R$ for some spatial rotation R.

$$\rightarrow \rho(A)^{-1} V_{Ap} = \pm V_p \rho(R). \qquad (*)$$

$$\therefore \underbrace{\rho(A)^\dagger V_{Ap}^{-2} \rho(A)}_{\text{positive-definite}} = \overbrace{V_p^{-1} (\rho(R)^{-1})^\dagger}^{\text{Inverse + dagger } (*)} \overbrace{\rho(R)^{-1} V_p^{-1}}^{\text{Inverse } (*)} = \pm V_p^{-2}$$

$(x^\dagger \rho(A)^\dagger V_{Ap}^{-2} \rho(A) x = (\rho(A)x)^\dagger V_{Ap}^{-2} (\rho(A)x) > 0)$

$\rho(R)\rho(R)^\dagger = \pm \rho(RR^\dagger) = \pm I.$

$$\rho(A)^\dagger V_{Ap}^{-2} \rho(A) = V_p^{-2}.$$

///

25.4 Multiparticle state space. (Dr. Shim)

Lec. 6.

$\mathcal{H}$: the Hilbert space for a single fermionic particle

$$\mathcal{H}^{\otimes n}_{anti} = \left\{ \sum_{i_1,\cdots,i_n \geq 1} \alpha_{i_1,\cdots,i_n} e_{i_1} \otimes \cdots \otimes e_{i_n} : \forall \sigma \in S_n, \ \alpha_{i_{\sigma(1)},\cdots,i_{\sigma(n)}} = \mathrm{sgn}(\sigma)\, \alpha_{i_1,\cdots,i_n} \right\}$$

Alt

Note. $\alpha_{i_1,\cdots,i_n} = 0$ if $\exists j \neq k,\ i_j = i_k$.

(Consider $(j\,k) \in S_n$)

. basis-independent.

$$\mathcal{F}_0 = \bigoplus_{n=0}^{\infty} \mathcal{H}^{\otimes}_{anti} \underset{\text{dense}}{\subset} \mathcal{F}$$

$$\mathcal{F} = \{ (\psi_0, \psi_1, \cdots) : \psi_n \in \mathcal{H}^{\otimes n}_{anti},\ \sum_{n=0}^{\infty} \|\psi_n\|^2 < \infty \}$$

is the closure under the inner product

$$(\psi, \varphi) = \sum_{n=0}^{\infty} (\psi_n, \varphi_n) \qquad \begin{cases} \psi = (\psi_0, \psi_1, \cdots) \\ \varphi = (\varphi_0, \varphi_1, \cdots) \end{cases}$$

for Bosonic particle

$$\mathcal{H}^{\otimes n}_{sym} = \left\{ \sum_{i_1,\cdots,i_n \geq 1} \alpha_{i_1,\cdots,i_n} e_{i_1} \otimes \cdots \otimes e_{i_n} : \forall \sigma \in S_n, \ \alpha_{i_{\sigma(1)},\cdots,i_{\sigma(n)}} = \alpha_{i_1,\cdots,i_n} \right\}$$

$$\mathcal{B}_0 := \bigoplus_{n=0}^{\infty} \mathcal{H}^{\otimes n}_{sym} \underset{\text{dense}}{\subset} \mathcal{B}$$

$$\mathcal{B} = \{ (\psi_0, \psi_1, \cdots) : \psi_n \in \mathcal{H}^{\otimes n}_{sym},\ \sum_{n=0}^{\infty} \|\psi_n\|^2 < \infty \}$$

is the closure under the inner product

$$(\psi, \varphi) = \sum_{n=0}^{\infty} (\psi_n, \varphi_n).$$

We will take $\mathcal{H} = L^2(X_m, d\lambda_m, \mathbb{C}^2)$

26. The Dirac field.

26.1 A basis for fermionic Fock space

For an orthonormal basis $e_1, e_2, \cdots$ of $\mathcal{H}$, can form an orthonormal basis for $\mathcal{H}_{anti}^{\otimes n}$ from elements of the form

$$\frac{1}{\sqrt{n!}} \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma)\, e_{i_{\sigma(1)}} \otimes \cdots \otimes e_{i_{\sigma(n)}} = \sqrt{n!}\, e_{i_1} \wedge e_{i_2} \wedge \cdots \wedge e_{i_n}.$$

where $i_1 < i_2 < \cdots < i_n$. denote this by $|n_1, n_2, \cdots\rangle$, where

$$n_j = \begin{cases} 1 & \text{if } j \in \{i_1, \cdots, i_n\} \\ 0 & \text{o.w.} \end{cases}$$

ex) $n=3$, $i_1=1$, $i_2=2$, $i_3=3$

$$\frac{1}{\sqrt{3!}} \sum_{\sigma \in S_3} \mathrm{sgn}(\sigma)\, e_{i_{\sigma(1)}} \otimes e_{i_{\sigma(2)}} \otimes e_{i_{\sigma(3)}}$$

$$= \frac{1}{\sqrt{3!}} \left(e_1 \otimes e_2 \otimes e_3 - e_2 \otimes e_1 \otimes e_3 - e_1 \otimes e_3 \otimes e_2 - e_3 \otimes e_2 \otimes e_1 + e_3 \otimes e_1 \otimes e_2 + e_2 \otimes e_3 \otimes e_1 \right)$$

$$= |1, 1, 1, 0, 0, \cdots\rangle$$

$|n_1, n_2, \cdots\rangle$: n particles in states $e_{i_1}, \cdots, e_{i_n}$.

<2.

* Pauli exclusion principle : the antisymmetry force to $n_j \in \{0, 1\}$:
  "no two identical fermions can occupy the same state."

* $\{ |n_1, n_2, \cdots\rangle : n_i \in \{0,1\} \ \& \ \sum n_i = n \}$: an orthonormal basis for $\mathcal{H}_{anti}^{\otimes n}$

  $\{ |n_1, n_2, \cdots\rangle : n_i \in \{0,1\} \ \& \ \sum n_i < \infty \}$: an orthonormal basis for $\mathcal{F}$

26.2

$k \geq 1$. $\psi = |n_1, n_2, \cdots\rangle$

- the creation a_k^+

$$a_k^+ \psi = \begin{cases} 0 & \text{if } n_k = 1, \\ (-1)^m |n_1, \cdots, n_{k-1}, 1, n_{k+1}, \cdots\rangle & \text{if } n_k = 0 \end{cases}, \quad m := \#\{1 \leq i < k : n_i = 1\} = \sum_{i=1}^{k-1} n_i$$

$0 \rightarrow$ Zero of the vectorspace $\mathcal{F}$ (not the vacuum)

Ex) $a_k^+ |0, 0, \cdots\rangle = e_k$

- the annihilation a_k.

$$a_k \psi = \begin{cases} 0 & \text{if } n_k = 0 \\ (-1)^m |n_1, \cdots, n_{k-1}, 0, n_{k+1}, \cdots\rangle & \text{if } n_k \neq 0. \end{cases}$$

- anticommutation relations: $\{A, B\} = AB + BA$

$\{a_k, a_\ell^+\} = \delta_{k,\ell} 1$, $\{a_k^+, a_\ell^+\} = 0$, $\{a_k, a_\ell\} = 0$ for all $k, \ell \in \mathbb{N}$.

i) $k = \ell$,

$$\{a_k, a_\ell^+\}\psi = a_k a_\ell^+ \psi + a_\ell^+ a_k \psi = \begin{cases} \psi & n_k = 0 \\ \psi & n_k = 1 \end{cases}$$

$k \neq \ell$

$$\{a_k, a_\ell^+\}\psi = \begin{cases} 0 & n_k = 0, \ n_\ell = 0 \\ 0 & n_k = 0, \ n_\ell = 1 \\ 0 & n_k = 1, \ n_\ell = 0 \\ 0 & n_k = 1, \ n_\ell = 1 \end{cases}$$

a_k, a_ℓ^+

· Continuous version of a_k^+, a_k.

Define $A^+, A : \mathcal{H} \longrightarrow \mathcal{L}(\mathcal{F}_0, \mathcal{F})$: $f \in \mathcal{H}$, $f = \sum_{k=1}^{\infty} \alpha_k e_k$

$$A^+(f)\psi := \sum_k \alpha_k a_k^+ \psi , \qquad A(f)\psi := \sum_k \alpha_k^* a_k \psi .$$

* A, A^+: basis independent

* $\{A(\xi), A^+(\eta)\} = (\xi, \eta)\mathbb{1}$, $\{A^+(\xi), A^+(\eta)\} = 0$, $\{A(\xi), A(\eta)\} = 0$ for all $\xi, \eta \in \mathcal{H}$.

Similar to p.31

Now if U is a unitary operator on $\mathcal{H}$, U induces a unitary operator on all the tensor powers on $\mathcal{H}$ and hence a unitary operator $\mathcal{F}(U)$ on the full tensor algebra $\mathcal{F}(\mathcal{H}) = \bigoplus_0^\infty \mathcal{H}^{\otimes k}$, by the formula

$$\mathcal{F}(U)(u_1 \otimes \cdots \otimes u_k) = Uu_1 \otimes \cdots \otimes Uu_k .$$

Using the commutativity of $\mathcal{F}(U)$ (with the number operator N and with the projections onto the symmetric and antisymmetric subspaces) and some auxiliary function, we have

$$\underset{\mathcal{F}(U)}{\overset{\shortparallel}{U}} A(\xi) \underset{\mathcal{F}(U)^{-1}}{\overset{\shortparallel}{U^{-1}}} = A(U\xi) , \qquad UA(\xi)^+U^{-1} = A(U\xi)^+ \text{ for all } \xi \in \mathcal{H}.$$

$\left(\mathcal{F}(U)^{-1}(u_1 \otimes \cdots \otimes u_k) = U^{-1}u_1 \otimes \cdots \otimes U^{-1}u_k \right)$ See Lec. 13 covered by Dr. Bae.

26.3 free field associated to electrons.

$\mathcal{H} = L^2(X_m, d\lambda_m, \mathbb{C}^2)$, $\mathcal{F}$: the fermion Fock space cor. to $\mathcal{H}$, e_1, e_2 : the standard basis of $\mathbb{C}^2$.

$p \in X_m$, $s \in \{1, 2\}$. For $\xi \in L^2(X_m, d\lambda_m)$

$$A^\dagger(\xi e_s) = \int_{X_m} d\lambda_m(p)\, \xi(p)\, a^\dagger(p, s) \quad , \quad A(\xi e_s) = \int_{X_m} d\lambda_m(p)\, \xi(p)^* a(p, s)$$

formally, $a^\dagger(p,s) = A^\dagger(\delta_p e_s)$, $a(p,s) = A(\delta_p e_s)$.

Recall that $X_m = \{ p \in \mathbb{R}^{1,3} \mid p^2 = m^2, p_0 \geq 0 \} = \{ (\omega_{\mathbf{p}}, \mathbf{p}) \ , \ \mathbf{p} \in \mathbb{R}^3, \ \omega_{\mathbf{p}} = \sqrt{m^2 + \mathbf{p}^2} \}$

define

$$a^\dagger(\mathbf{p}, s) := \frac{a^\dagger(p,s)}{\sqrt{2\omega_{\mathbf{p}}}} \quad , \quad a(\mathbf{p}, s) = \frac{a(p,s)}{\sqrt{2\omega_{\mathbf{p}}}} ,$$

so that

$$\{ a(\mathbf{p}, s) , a^\dagger(\mathbf{q}, t) \} = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q})\, \delta_{s,t}\, 1 \quad , \quad \{ a(\mathbf{p}, s), a(\mathbf{q}, t) \} = 0 = \{ a^\dagger(\mathbf{p}, s), a^\dagger(\mathbf{q}, t) \}$$

$\because$ $A^\dagger(\xi e_s) = \int_{X_m} d\lambda_m(p)\, \xi(p) \sqrt{2\omega_{\mathbf{p}}}\, a^\dagger(\mathbf{p}, s) = \int_{\mathbb{R}^3} \frac{d\mathbf{p}^3}{(2\pi)^3} \cdot \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} \xi(\omega_{\mathbf{p}}, \mathbf{p})\, a^\dagger(\mathbf{p}, s)$ similar to p.48

$$\iint \frac{d\mathbf{p}^3 d\mathbf{q}^3}{(2\pi)^6} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} \frac{1}{\sqrt{2\omega_{\mathbf{q}}}} \xi(\mathbf{p}) \cdot \eta(\mathbf{q})^* \{ a(\mathbf{p}, s), a^\dagger(\mathbf{q}, t) \}$$

symbolic expression. (pp 31-32)

$$= \{ A(\xi e_s), A^\dagger(\eta e_{s'}) \} = (\xi e_s, \eta e_{s'}) 1 = \int \frac{d\mathbf{p}^3}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{p}}} \cdot \xi e_s(\mathbf{p})\, \eta^* e_{s'}(\mathbf{p})$$

$$= \iint \frac{d\mathbf{p}^3 d\mathbf{q}^3}{(2\pi)^6} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} \frac{1}{\sqrt{2\omega_{\mathbf{q}}}} (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q})\, \delta_{s,t}\, \xi(\mathbf{p})\, \eta(\mathbf{q})^*$$

for all ξ, η.

Now recall that $p^* = (m,0,0,0) \in X_m$, $\kappa : SL(2,\mathbb{C}) \to SO^\uparrow(1,3)$, and $\exists ! V_p$, $\kappa(V_p)p^* = p$.
Define $S_p \in GL(4,\mathbb{C})$ by

$$S_p := \begin{pmatrix} (V_p^+)^{-1} & 0 \\ 0 & V_p \end{pmatrix}.$$

Define a basis of $\mathbb{C}^4$:

$$f_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix},\quad f_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix},\quad f_1' = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix},\quad f_2' = \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}.$$

Define

$$u(p,s) := S_p f_s = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} \quad \text{and} \quad v(p,s) := S_p f_s' = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix}.$$

Let $\mathcal{G} = \mathcal{F} \otimes \mathcal{F}'$ where $\mathcal{F}, \mathcal{F}'$ are two copies of the Fermion Fock space for electrons, which denote the joint electron-positron space.

antiparticle counterpart of the electron. (same mass and spin but electric charge of $+1e$)

Let b^+, b on $\mathcal{F}'$ $\sim$ a^+, a on $\mathcal{F}$.

Note $\forall C : \mathcal{F} \circlearrowleft$ extends uniquely to $\mathcal{G}$ via $C(f \otimes f') = Cf \otimes f'$.

For $k \in \{1,2,3,4\}$, we define an operator-valued distribution ψ_k on $\mathbb{R}^{1,3}$ as

$$\psi_k(x) := \sum_{s=1}^{2} \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3} \frac{\sqrt{m}}{\sqrt{2\omega_p}} \left[e^{-i(x,p)} u_k(p,s)\, \underline{a(p,s)} + e^{i(x,p)} v_k(p,s)\, \underline{b^+(p,s)} \right].$$

$$\psi(x) := \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \\ \psi_3(x) \\ \psi_4(x) \end{pmatrix}$$: the (quantized) Dirac field. (Dirac spinor field)

Compare φ in p.51.

Further see [1] Chap. 5.